

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

C02: Apply suitable Data Pre processing techniques like Data cleaning, Data integration, Data transformation and data Reduction, for effective data preparation (BL3, BL4)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding (2)	Method Selection (2)	Solution Process (2.5)	Accuracy of Calculation (2)	Interpretation of Results (1.5)	Total(10)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I **C.Nandeesh(192525015)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process

		approach throughout			
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Analytical problems

- ① Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order).

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25
25, 30, 33, 33, 35, 35, 35, 35, 35, 36, 40, 45
46, 52, 70.

- ② what is the mean of data & median.
③ what is the mode of the data
comment on the data's modality
(i.e., bimodal, trimodal, etc).
④ what is the midrange of the data.
⑤ can you find the first Quartile (Q_1)
 Q_3
⑥ give the five-number summary of
the data.
⑦ show a boxplot of the data.

$$a) \frac{\sum x}{N}$$

Mean

$$13+15+16+16+17+20+20$$

$$\frac{809}{27} = 29.96$$

Median

$$n=27$$

$$\frac{8+11}{2} = \frac{28}{2} = 14^{\text{th}} \text{ value.}$$

$$\boxed{\text{Median} = 25}$$

b) mode & modality

$$\text{mode} = 25, 35$$

$$\boxed{\text{modality} = bimodal}$$

c)

$$\text{range} =$$

$$= \frac{\text{Max} + \text{Min}}{2}$$

$$= \frac{70+13}{2}$$

$$= \frac{83}{2} = 41.5$$

①

24 values.

Q_1 position

$Q_1 = 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25$

median = $\frac{14}{2} = 7^{th}$ value.

$Q_1 = 20$

$Q_3 = 30, 33, 33, 35, 35, 35, 35, 35, 35, 36, 40, 45$
46, 52, 70.

= $\frac{14+1}{3} = \frac{15}{3} = 5^{th}$ value.

$Q_3 = 35$

②

1. $M_{99} = 13$

2. $Q_1 = 20$

3. median = 25

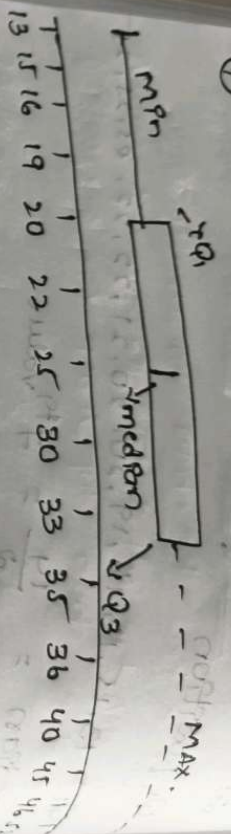
4. $Q_3 = 35$

5. $Max = 70$

SS = 100511



① box plot



② given the following measurements

for the variable age:

18, 22, 25, 42, 28, 43, 33, 35, 16, 08

compute the mean absolute deviation of age.

$$\text{Sol: } \text{mean} = \frac{\sum x}{N}$$

$$= \frac{330}{10}$$

$$\boxed{\text{mean} = 33}$$

x	$ x - \bar{x} $	f
18	$ 18 - 33 $	15
22	$ 22 - 33 $	11
25	$ 25 - 33 $	8
42	$ 42 - 33 $	9
28	$ 28 - 33 $	5
43	$ 43 - 33 $	10
33	$ 33 - 33 $	0
35	$ 35 - 33 $	2
56	$ 56 - 33 $	23
28	$ 28 - 33 $	5

$$MAD = \frac{\sum |x - \bar{x}|}{n}$$

$$\frac{88}{10} = 8.8$$

years.

3) Suppose that the data for analysis includes the attribute age. The age values for the data tuples are

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70

use smoothing by bin means to smooth the above data, using a bin depth

of 3.

$$\text{mean} = \frac{\sum x}{N}$$

$$N = 87$$

$$\text{bwn depth} = 3$$

$$\text{count of bwn} = \frac{\text{Accuracy}}{\text{bwn depth}}$$

$$= \frac{27}{3}$$

$$= 9$$

B₁ original values

mean

13, 15, 16

$$\frac{13+15+16}{3} = 14.67$$

B₂

16, 19, 20

$$\frac{16+19+20}{3} = 18.33$$

B₃

20, 21, 22

21

B₄

22, 25, 25

24

B₅

25, 25, 30

26.67

B₆

33, 33, 35

33.67

B₇

35, 35, 35

35

B8 | ~~42, 46~~ 20, 36, 40, 45 | ~~34~~ 46, 33
 B9 | ~~40~~ 46, 52, 70 | ~~48~~ 56

④ use the two methods below to normalize the following data.

200, 300, 400, 600, 1000

⑤ Min-Max normalization.

min = 0 & Max = 1

⑥ Z-score normalization.

Sol:-

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$x' = \frac{x - 200}{1000 - 200}$$

$$x' = \frac{x - 200}{800} = \frac{x - \min}{\max - \min}$$

x	x'
200	$\frac{200 - 200}{800} = 0$
300	$\frac{300 - 200}{800} = \frac{100}{800} = \frac{1}{8} = 0.125$
400	$\frac{400 - 200}{800} = \frac{200}{800} = \frac{1}{4} = 0.25$
600	$\frac{600 - 200}{800} = \frac{400}{800} = \frac{1}{2} = 0.5$
1000	$\frac{1000 - 200}{800} = \frac{800}{800} = 1$

$$\text{mean} = \frac{0.30 + 1.5 + 1.0 + 1.0 + 1.0}{5}$$

$$= \frac{5.8}{5} = 1.16$$

$$\mu = 500$$

x	$x - \mu$	$(x - \mu)^2$
200	-300	90000
300	-200	40000
400	-100	10000
600	100	10000
1000	500	250000

$$\sum (x - \mu)^2 = 400000$$

$$\sigma^2 = \frac{400000}{5} = 80000$$

$$\sigma^2 = 80000$$

$$\sigma = \sqrt{80000}$$

$$= 282.84$$

orig value		zscore.
200	$\frac{-200}{282.84}$	-1.061
300	$\frac{-100}{282.84}$	-0.707
400	$\frac{0}{282.84}$	0
600	$\frac{100}{282.84}$	0.354
1000	$\frac{500}{282.84}$	1.768

$$zscore = -1.061, -0.707, 0, 0.354, 1.768$$

⑤ Suppose a group of 12 sales price records has been sorted as follows:
 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 815.
 partition them into three bins by each of the following methods.

- equal-frequency partitioning
- equal-width partitioning
- clustering.

⑥ $\frac{12}{3} = 4 \text{ records.}$

values

B₁ 5, 10, 11, 13

B₂ 15, 35, 50, 55

B₃ 72, 92, 204, 215

(b)
$$\text{width} = \frac{\text{max} - \text{min}}{3}$$

$$\text{width} = \frac{805 - 5}{3} = \frac{800}{3} = 266.67 \approx 267$$

$$\text{min} = 70$$

- B₁ 5-275
- B₂ 75-145
- B₃ 145-215



③

clusters

c_1

c_2

c_3

values

5, 10, 11, 13, 15

85, 50, 55, 71, 92

204, 1215

⑥

~~uh~~
~~25/7/20~~

age	22	22	27	27	27	41	47	49	50
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